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### Tracking of retinal traction through digital image correlation

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#### Abstract

A tracking system for quantifying retinal traction on a retina of a patient eye includes an indicator device, a stereo camera, and an electronic control unit (ECU). The stereo camera collects and outputs stereo image data. The ECU, which is in communication with the indicator device, executes a method by receiving the stereo image data from the stereo camera during an ophthalmic procedure and thereafter assigning tracking points as coinciding pixels within stereo image pairs. The ECU also automatically performs a digital image correlation (DIC) process using the stereo image pairs to ascertain relative motion of the tracking points, and associates relative motion of the tracking points with the retinal traction using a traction map. A numeric traction quotient is generated that is indicative of magnitude of retinal traction. The ECU executes a control action using the indicator device based on the numeric traction quotient.

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## Field of Classification Search

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## Background/Summary

**CROSS-REFERENCE TO RELATED APPLICATIONS** (1) This application is a continuation of U.S. Nonprovisional patent application Ser. No. 17/490,456 filed Sep. 30, 2021, which claims priority to and the benefit of U.S. Patent Application Ser. No. 63/151,954 filed on Feb. 22, 2021, both of which are herein incorporated by reference in their respective entireties.

### TECHNICAL FIELD

(1) The present disclosure relates to an imaging-based strategy for quantifying, tracking, and mitigating tractive forces on a retina during retinal surgery.

### BACKGROUND

(2) Vitrectomy and other invasive surgeries of the eye require a surgeon to insert specialized surgical tools into the vitreous cavity of a patient's eyeball, and to thereafter manipulate the surgical tools when performing a particular surgical technique. The surgeon's actions are guided in real-time by highly magnified imagery of the retina and surrounding intraocular tissue. To this end, magnified retinal images are typically displayed within view of the surgeon and other attending clinicians. At the same time, the magnified retina may be viewed in other ways, such as through eye pieces of a high-power ophthalmic microscope.

(3) As appreciated in the art, the vitreous cavity extends between the lens and the retina of the human eye, with the lens and the retina being respectively located at the anterior region and the posterior region of the eyeball. The vitreous cavity is occupied by a transparent, gel-like substance referred to as the vitreous humor, which itself is encapsulated within a thin membrane called the

vitreous cortex. The retina is isolated from the vitreous body by another thin intervening layer of tissue, i.e., the inner limiting membrane or ILM.

(4) Due to adherence of the ILM to the retinal surface, common ophthalmic surgeries such as the repair of macular holes or torn retinas, removal of scar tissue, and other delicate surgeries of the eye may require the attending surgeon to securely grasp and carefully peel the ILM away from the backing retina. ILM peeling has the effect of relaxing the disturbed retina, while also providing the surgeon with unimpeded access to the retinal surface. Performance of an ILM peeling procedure involves the deliberate application of surgeon-imparted tractive forces to the ILM. As the surgeon maneuvers the ILM away from the retina, such forces are transferred to the attached retina. The resulting traction across the retinal surface is referred to herein and in the general art as retinal traction.

#### SUMMARY

(5) Disclosed herein are automated imaging-based systems and methods for quantifying retinal traction during the performance of an ophthalmic procedure, principally but not necessarily limited to vitrectomy and other invasive eye surgeries. As appreciated in the art, alternative approaches for manipulating the inner limiting membrane (ILM) include a forceps-assisted “pinch-and-peel” technique and a friction-based technique, the latter of which utilizes a specialized scraping loop, e.g., the FINESSE™ Flex Loop Curved Nitinol loop commercially from Alcon, Inc. Use of both exemplary tools results in retinal traction, which may lead to iatrogenic tearing of the ILM and other possible eye trauma.

(6) ILM structural integrity for a given patient eye tends to vary due to factors such as heredity, age, injury, and disease. As a result, the effect of a particular magnitude and duration of applied retinal traction on a given patient eye is difficult to predict. Likewise, variations in surgeon skill level, intrinsic capabilities and limitations of the surgical tools employed by a given surgeon, and other factors can produce widely different end results. The present solutions are therefore intended to reduce uncertainty during retinal surgeries, while at the same time improving surgical outcomes and increasing overall surgeon confidence.

(7) In order to accomplish these and other possible goals, the present approach relies on collection and processing of stereo images of the retina, along with an automated and/or surgeon-directed assignment of target pixels (“tracking points”) as coinciding pixels within stereo image pairs of the stereo image data. Relative motion of the assigned tracking points is closely monitored during the ophthalmic procedure using an electronic control unit (ECU), e.g., a standalone or distributed computer device, or associated hardware integrated partially or fully with a stereo camera in different embodiments.

(8) The ECU described herein uses a traction model to automatically quantify retinal traction, with the ECU ultimately outputting a numeric traction quotient indicative of a magnitude of such retinal traction. In a simplified approach, the numeric traction quotient may be a unitless value indicative of the magnitude, e.g., a normalized value having a maximum value of 1 and a minimum value of 0. The ECU automatically alerts the surgeon in real-time based on the numeric traction quotient, such as when the numeric traction quotient, possibly averaged across the entire surface area of the retina or within designated zones or regions thereof, exceeds a corresponding traction threshold. Real-time audio, visual, and/or tactile alerts may be generated in some embodiments to enable the surgeon to make more informed adjustments to tugging or pulling forces imparted by the surgeon to the ILM.

(9) In a non-limiting exemplary embodiment, a tracking system for quantifying such retinal traction during an ophthalmic procedure includes an indicator device, a stereo camera, and an ECU. The ECU, which is in wired or wireless communication with the indicator device in this particular embodiment, is configured to receive stereo image data from the stereo camera, with the ECU possibly integrated with the stereo camera or existing as separate hardware in communication therewith. The controller thereafter assigns target pixels within the stereo image pairs,

autonomously or using surgeon-directed input signals, with such assigned target pixels hereinafter referred to as “tracking points” for simplicity and clarity.

(10) The ECU in this representative configuration is programmed with the above-noted traction model. While various approaches could be used to implement the envisioned traction model, one possible solution includes a software-based logic block configured to automatically perform a digital image correlation (DIC) process to the collected stereo images, with the DIC process used to ascertain relative motion of the tracking points. The ECU then associates the relative motion of the tracking points with the particular retinal traction causing such motion to occur, e.g., using a lookup table, with the ECU thereafter outputting the above-noted numeric traction quotient. The provided numeric traction quotient is thus indicative of the magnitude of the retinal traction as noted above. The ECU then executes an appropriate control action with respect to the indicator device based on the numeric traction quotient, for instance when the magnitude of the numeric traction quotient exceeds a pre-calibrated threshold or a user-calibratable threshold.

(11) In order to ensure that the relative motion tracked by the ECU results principally from retinal traction applied by the surgeon, as opposed to baseline motion caused by other forces such as patient-induced and/or externally-induced eye motion, the ECU may be configured to apply a free-body/solid-body motion filter to the relative motion to account for and ultimately filter out such baseline motion.

(12) The indicator device may include one or more high-resolution display screens, for instance 4K or higher resolution LED-backlit medical-grade monitors. In such an embodiment, the ECU may be configured to automatically present an intuitive “heat map” of the retinal surface via the display screen(s), alone or in conjunction with text messages or prompts. The displayed heat map, which is graphically representative of the present level of retinal traction, may pinpoint locations of relatively high traction in some configurations, e.g., on a pixel-by-pixel basis or a region-by-region basis. Such a heat map could be displayed on top of a displayed stereo image of the retina, such as in the form of an overlay to a three-dimensional fundus image, to indicate corresponding high-traction zones.

(13) An accompanying method is also disclosed for quantifying retinal traction during an ophthalmic procedure. According to an exemplary embodiment, the method includes receiving the stereo image data from the stereo camera via the ECU during an ophthalmic procedure, with the stereo image data including a stereo image pair. The method also includes assigning tracking points as coinciding pixels within the stereo image pair, and automatically performing a DIC process, via the ECU, using the stereo image pair. In this manner the ECU ascertains relative motion of the tracking points.

(14) Furthermore, the method in this particular embodiment includes associating relative motion of the tracking points with the retinal traction, using a traction map of the ECU, to thereby determine a numeric traction quotient indicative of magnitude of the retinal traction. The ECU thereafter executes a control action using the indicator device, with the control action being based on the numeric traction quotient and indicative of the magnitude of the retinal traction.

(15) A system is also disclosed herein for quantifying retinal traction on a retina of a patient eye, with the system including a central processing unit (CPU) and computer-readable media on which is recorded a set of instructions. Execution of the instructions by the CPU causes the CPU to receive stereo image data, inclusive of a stereo image pair or pairs, from a stereo camera during the ophthalmic procedure, and to assign tracking points as coinciding pixels within the stereo image pair(s). The CPU also automatically performs a DIC process using the stereo images to ascertain relative motion of the tracking points. In this particular embodiment, the CPU associates relative motion of the tracking points with the retinal traction, using a traction map, as a numeric traction quotient indicative of magnitude of the retinal traction. The CPU thereafter communicates a control signal to an external indicator device when the numeric traction quotient exceeds one or more calibrated traction thresholds.

(16) The above-described features and advantages and other possible features and advantages of the present disclosure will be apparent from the following detailed description of the best modes for carrying out the disclosure when taken in connection with the accompanying drawings.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic illustration of an operating room setup using a tracking system for automatically quantifying and tracking retinal traction during a representative ophthalmic procedure using a digital image correlation (DIC) process.
- (2) FIG. 2 is a schematic depiction of a representative ophthalmic procedure in which an inner limiting membrane (ILM) is grasped and peeled away from a retina, thereby imparting traction forces to the retina.
- (3) FIG. 3 is an exemplary embodiment of the tracking system shown in FIG. 1.
- (4) FIG. 4 is a schematic illustration of a heat map depicting areas or zones of elevated retinal traction in accordance with an aspect of the disclosure.
- (5) FIG. 5 is a schematic illustration of stereo image data and associated tracking points usable as part of the disclosed solutions.
- (6) FIG. 6 is a flow chart describing an exemplary method for quantifying and tracking retinal traction using the traction system shown in FIG. 1.
- (7) The foregoing and other features of the present disclosure are more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings.

### DETAILED DESCRIPTION

(8) Embodiments of the present disclosure are described herein. It is to be understood, however, that the disclosed embodiments are merely examples and other embodiments can take various and alternative forms. The figures are not necessarily to scale. Some features could be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present disclosure. As those of ordinary skill in the art will understand, various features illustrated and described with reference to any one of the figures can be combined with features illustrated in one or more other figures to produce embodiments that are not explicitly illustrated or described. The combinations of features illustrated provide representative embodiments for typical applications. Various combinations and modifications of the features consistent with the teachings of this disclosure, however, could be desired for particular applications or implementations.

(9) Certain terminology may be used in the following description for the purpose of reference only, and thus are not intended to be limiting. For example, terms such as “above” and “below” refer to directions in the drawings to which reference is made. Terms such as “front,” “back,” “fore,” “aft,” “left,” “right,” “rear,” and “side” describe the orientation and/or location of portions of the components or elements within a consistent but arbitrary frame of reference which is made clear by reference to the text and the associated drawings describing the components or elements under discussion. Moreover, terms such as “first,” “second,” “third,” and so on may be used to describe separate components. Such terminology may include the words specifically mentioned above, derivatives thereof, and words of similar import.

(10) Referring to the drawings, wherein like reference numbers refer to like components, a representative surgical suite **10** is depicted schematically in FIG. 1. Such a surgical suite **10** may be equipped with a multi-axis surgical robot **12** and an operating platform **14**, e.g., a table as shown or an adjustable/reclinable surgical chair. When the surgical suite **10** is occupied by a surgical team (not shown), the multi-axis surgical robot **12** may be connected to a stereo camera **16** through

which a patient's intraocular anatomy may be visualized in three dimensions under high magnification, as appreciated in the art. Using associated hardware and software, a surgeon, in a heads-up manner using three-dimensional (3D) viewing glasses (not shown), is able to accurately visualize targeted tissue using highly magnified 3D images **18** and **118** of a retina **25** and surrounding anatomy, which may be displayed or projected using corresponding high-resolution medical display screens **20** and/or **200**. Such display screens **20** and **200** are one possible embodiment of an indicator device (IND) **20A** as shown in FIG. **3**, enabling heads up viewing by the surgeon. Heads up viewing in this manner has the benefit of reducing stress and strain on the surgeon's neck and back relative to conventional top-down viewing of targeted tissue through eye pieces of an ophthalmic microscope.

(11) The stereo camera **16**, which may be configured as shown in the exemplary embodiment of FIG. **1** or in various other application-suitable sizes and/or shapes, includes or is communicatively connected to local control processor (LCP) **36**. The LCP **36** may be embodied as a microprocessor, an application-specific integrated circuit (ASIC), central processor unit, etc., and is configured to collect and output stereo image data **38**. That is, for each instant in time according to a predetermined sampling interval, two digital images are concurrently collected as stereo image pair **38P** (Image **1**, Image **2**). As appreciated in the art, when the stereo image pair **38P** is viewed by the surgeon and other attending clinicians through 3D glasses, the stereo image pair **38P** converges into a 3D image.

(12) Surgically useful levels of optical and/or digital magnification, and digital imaging of the retina **25** and other intraocular anatomy, is enabled by the ever-evolving capabilities of modern ophthalmic/surgical quality optical devices. The stereo camera **16** is one such device. The real-time availability of the stereo image data **38** from the stereo camera **16** during vitrectomy or other ophthalmic procedures is thus an enabling technology for the present solution described below with reference to FIGS. **2-6**. Relative to approaches using a conventional ophthalmic microscope, during which the surgeon views the retina **25** through optical eye pieces throughout the surgery, the present solution allows the surgeon to maintain a more ergonomically friendly “heads up” posture, thus reducing stress and strain on the surgeon's neck and back.

(13) Also present within the exemplary surgical suite **10** of FIG. **1** is a cabinet **22** containing an electronic control unit (ECU) **50** in communication with the indicator device **20A** of FIG. **3**. In different possible implementations, computer-readable instructions embodying a method **70**, exemplified in FIG. **6**, may reside aboard the ECU **50**. The ECU **50** in turn may be a standalone computer as show in FIG. **1**, a distributed/networked system, or partially or fully integrated with the stereo camera **16**. The cabinet **22**, depicted in an optional location as being collocated with the display screen **20**, may be positioned elsewhere in the surgical suite **10**. Such a cabinet **22** may be constructed of a lightweight and easily sanitized construction, such as painted aluminum or stainless steel, and used to protect constituent hardware from possible ingress of dust, debris, and moisture. For improved visibility, light may be emitted by a lamp **17** mounted to an optical head of the stereo camera **16**, and possibly from an endoilluminator **32** inserted into the patient's eye **30** of FIG. **3**.

(14) The ECU **50** of FIG. **1** is configured to receive the stereo image data **38**, i.e., sequential stereo image pairs **38P** represented schematically by arrows Image **1** and Image **2**, with the ECU **50** receiving such stereo image data **38** from the stereo camera **16** in real-time. As part of the method **70**, the ECU **50** assigns tracking points as coinciding pixels within the stereo image pair **38P**, autonomously or with the involvement of the surgeon, and then automatically performs a digital image correlation (DIC) process on the stereo image pairs **38P** to ascertain relative motion of the tracking points. Additionally, the ECU **50** associates relative motion of the tracking points with the retinal traction, such as by using a traction map **55** as depicted in FIG. **3**, with the ECU **50** doing so as a numeric traction quotient indicative of magnitude of the retinal traction. The ECU **50** then executes a control action with respect to the indicator device(s) **20A** and/or **20B** of FIG. **3** and/or

other audio, visual, or tactile devices based on the numeric traction quotient.

(15) Referring briefly to FIG. 2, a representative patient eye **30** is shown undergoing an inner limiting membrane (ILM) peeling procedure in which an ILM **31** is carefully separated and peeled away from the backing retina **25** using a surgical tool **34**. To accomplish the ILM peel, the surgeon carefully manipulates the surgical tool **34** to impart traction forces (arrow F.sub.T) to the ILM **31**. In this manner, the surgeon is able to expose the retina **25** in preparation for work thereon. In actual practice the ILM **31**, being thin, flexible, and transparent, is not easily identifiable. It is therefore common for the surgeon to apply a small amount of contrast-enhancing staining dye to the ILM **31**, such as indocyanine green (ICG) or Membrane-Blue-Dual (DORC International), to lightly stain the ILM **31** and thereby improve contrast. With visualization of the ILM **31** enhanced in this manner, the surgeon may commence peeling the ILM **31** to expose the retina **25**, as appreciated in the art.

(16) As noted above, the structural integrity of the ILM **31** of a given patient is expected to vary, possibly quite widely, due to factors such as heredity, age, injury, and disease. The effects of retinal traction during a particular surgical instance are therefore difficult to predict absent the present teachings. Likewise, variations in surgical skill and the intrinsic capabilities and particular choice of surgical tools can collectively produce different surgical results over time. The present solutions are therefore intended to facilitate real-time monitoring of retinal traction during ILM peeling and other procedures, while also providing the surgeon with real-time intuitive feedback. Together, the monitoring and intuitive feedback allow the surgeon, whether working alone or with the assistance of the exemplary surgical robot **60** of FIG. 3, to make any necessary adjustments for optimal surgical results.

(17) Referring now to FIG. 3, the patient eye **30** is shown undergoing a representative ophthalmic procedure **13**, in this instance an invasive vitreoretinal surgery, during operation of a tracking system **100** constructed as set forth in detail herein. The tracking system **100** is operable for quantifying retinal traction within the patient eye **30**. To this end, the tracking system **100** includes the indicator devices **20A** and **20B**, the stereo camera **16**, and the ECU **50**, the latter two of which may be integrated into a unitary device in some embodiments. During the course of the ophthalmic procedure **13**, the endoilluminator **32** may be inserted into a vitreous cavity **15** of the patient eye **30**. Light LL emitted from a distal end E1 of the endoilluminator **32**, as well as some light from the lamp **17** of FIG. 1, is used to illuminate the vitreous cavity **15**. Various lighting technologies may be used to emit the light LL, such as but not limited to red/green/blue (RGB) lasers, light-emitting diodes (LEDs), halogen bulbs, etc.

(18) During the representative ophthalmic procedure **13** of FIG. 3, the surgeon may be required to insert the surgical tool **34** into the vitreous cavity **15** in order to perform an operating task on or in proximity to the retina **25**. As appreciated by those skilled in the art, traditional approaches for manipulating the ILM **31** shown in FIG. 2 include a forceps-assisted “pinch-and-peel” technique, or the alternative use of a specialized scraping tool, such as the FINESSE™ Flex Loop Curved Nitinol loop from Alcon, Inc. Thus, the surgical tool **34** within the scope of the present disclosure may encompass either or both devices.

(19) The above techniques are typically performed manually using the dexterous skill of the surgeon to manipulate the surgical tool **34**. However, evolving machine vision-assisted robotic surgical techniques enable construction in some embodiments of a semi-automated or automated peeling procedure, e.g., using a multi-axis surgical robot **60**. For instance, the surgical tool **34** could be attached to an end-effector **60E** of the surgical robot **60**, which in turn may be placed in communication with the ECU **50**. Such a surgical robot **60** could be teleoperated by the surgeon through interaction with the ECU **50**, either directly or via a human-machine interface (not shown) such as a surgical workstation. Alternatively, the surgical robot **60** could have limited support functionality, such as by offloading stress from the surgeon by supporting the weight of the surgical tool **34**, reducing instances of tremors, etc., while leaving manipulation actions solely to the

surgeon. Such automated peeling techniques, executed or supported to some extent by the surgical robot **60**, would require in-depth knowledge of retinal traction, such as when measuring a retina indentation force as appreciated in the art.

(20) With respect to the endoilluminator **32**, the directed light LL falls incident upon exposed surfaces of the retina **25** to produce an illuminated retina surface **25I**, inclusive of the ILM **31** of FIG. 2, which is attached to the retina **25**. The endoilluminator **32** is coupled to an accompanying power supply (PS) **37**, controllable via a lighting control signals (arrow CC.sub.L) from the ECU **50** or another control processor, such as a filtered wall outlet or a battery pack and power inverter suitable for ensuring reliable generation and transmission of the directed light (arrow LL). During the course of the ophthalmic procedure **13**, the stereo camera **16** collects the stereo image data **38** of the illuminated retina surface **25I** and ILM **31** (FIG. 2), and thereafter transmits the collected stereo image data **38** to the ECU **50** for processing in accordance with the present retinal traction (R-TRAC) method **70**.

(21) The indicator device (IND) **20A**, e.g., the display screens **20** and/or **200** of FIG. 1, is likewise in communication with the ECU **50**, and configured to activate/turn on in response to an indicator control signal (arrow CC.sub.20A) from the ECU **50**. In response to the indicator control signal (arrow CC.sub.20A), and depending on the particular configuration of the indicator device **20A**, the indicator device **20A** may provide a suitable visual alarm, such as the heat map **45** shown in FIG. 4 and described below. The ECU **50** thus uses the indicator device **20A** to present an intuitive graphical depiction of retinal traction levels relative to the surface area of the retina **25**.

(22) Another similarly configured indicator device **20B** may be used in conjunction with the indicator device **20A** to provide multiple alerts, perhaps escalating alerts in response to results of the method **70**. For instance, the indicator device **20B** may provide audio, visual, and/or tactile alarms or warnings. A possible implementation of the indicator device **20B** is that of an audio speaker, in which case an indicator control signal (arrow CC.sub.20B) may cause the indicator device **20B** to sound an audible chime or warning tone. Alternatively, the indicator device **20B** may include a color-coded lamp, such that receipt of the indicator control signal (arrow CC.sub.20B) causes the indicator device **20B** to light up in a readily identifiable manner, e.g., using red light. Tactile feedback such as low-level vibration may be presented to the surgeon or another clinician in the surgical suite **10** of FIG. 1, with possibilities including a wearable device, floor mat, etc.

(23) Although the ECU **50** of FIG. 3 is depicted schematically as a unitary box for illustrative clarity and simplicity, the ECU **50** could include one or more networked devices each with a central processing unit (CPU) **52** and sufficient amounts of memory **54**, i.e., computer-readable media, including a non-transitory (e.g., tangible) medium that participates in providing data/instructions that may be read by the CPU **52**. Instructions embodying the method **70** and the accompanying traction map **55** may be stored in memory **54** and executed by the CPU **52** to cause the CPU **52** to perform the various functions described herein, thus enabling the present method **70**, possibly in conjunction with calibratable inputs **58** (arrow CAL).

(24) The memory **54** may take many forms, including but not limited to non-volatile media and volatile media. As understood in the art, non-volatile media may include optical and/or magnetic disks or other persistent memory, while volatile media may include dynamic random-access memory (DRAM), static RAM (SRAM), etc., any or all which may constitute a main memory of the ECU **50**. Input/output (I/O) circuitry **56** may be used to facilitate connection to and communication with the various peripheral devices used during the ophthalmic procedure **13**, inclusive of the stereo camera **16**, the endoilluminator **32**, and the indicator devices **20A** and/or **20B**. Other hardware not depicted but commonly used in the art may be included as part of the ECU **50**, including but not limited to a local oscillator or high-speed clock, signal buffers, filters, etc.

(25) Referring briefly to FIG. 4, the retina **25** is shown as a representative fundus image **42**. As appreciated in the art, the fundus image **42** is typically embodied as a color, black and white, or

grayscale image of various key structure of the retina **25**, primarily the optic disc **44**, the retinal artery **46** and surrounding veins stemming therefrom, and the macula **48**. The fundus image **42**, being ubiquitous in ophthalmic practice and thus familiar to attending clinicians, may be used as an intuitive backdrop to the displayed heat map **45**. In such a configuration, the ECU **50** may be configured to digitally divide or otherwise separate the retina **25** into multiple virtual zones, with two such zones **Z1** and **Z2** depicted in FIG. **4**, and to map retinal traction to the retina surface **25**. In this manner, the multiple zones may have a corresponding level of retinal traction that could be separately diagnosed and responded to as an alternative to, e.g., averaging retinal traction across the entire surface area of the retina **25** and applying a single traction threshold.

(26) In the optional display configuration of FIG. **4**, the ECU **50** could overlay the heat map **45** onto the fundus image **42**, i.e., a stereo image formed by the stereo image pair **38P** of FIG. **1**, with this information presented in real-time via the indicator device(s) **20A** and/or **20B**. The heat map **45** thus intuitively provides information that, at a glance, is indicative of a distribution or concentration of a magnitude of retinal traction across the retina **25**.

(27) Referring now to FIG. **6**, execution of instructions stored or recorded in the memory **54** of the ECU **50** as shown in FIG. **3** causes the CPU **52** and other hardware of the ECU **50** to perform the method **70**. A representative embodiment of the method **70** commences with logic block **B72**, which includes illuminating the retina **25** of FIG. **2**, e.g., with directed light **LL** from the endoilluminator **32** of FIG. **3**, along with possibly some additional light from the lamp **17** of FIG. **1**. Surgical steps preceding implementation of logic block **B72** would include creating an incision in the patient eye **30** of FIG. **3** and carefully inserting the endoilluminator **32** and the surgical tool **34** into the vitreous cavity **15**. The method **70** then proceeds to logic block **B74**.

(28) Logic block **B74** of FIG. **6** entails receiving the stereo image data **38** of FIG. **3** from the stereo camera **16** via the ECU **50**. This occurs in real-time during the ophthalmic procedure **13**. The ECU **50** may be separate from the stereo camera **16** in some embodiments, as depicted for illustrative clarity. Alternatively, the ECU **50** may include the LCP **36** of the stereo camera **16**, such that the LCP **36** is integral with the ECU **50**, i.e., the ECU **50** and the stereo camera **16** are effectively a single functional unit, thus reducing component count and possibly providing other processing efficiencies and reduced communications latency.

(29) Referring briefly to FIG. **5**, the collected stereo image data **38** is formed in digital embodiments from image pixels, and therefore logic block **B74** of FIG. **6** also includes identifying a corresponding pixel field of each of the constituent images of the stereo image pair **38P**. For example, Image **1** and Image **2** are shown as 8 pixel by 8 pixel (8×8) digital images arranged in a nominal (X, Y) pixel coordinate system. As Images **1** and **2** are collected at the same instant in time, an assigned pixel (4, 5) in Image **1** coincides with the same pixel (4, 5) in Image **2**, and so forth. That is, within a given stereo image pair **38P**, pixel (4, 5) of Image **1** coincides with pixel (4, 5) of Image **2**. As part of logic block **B74** of FIG. **6**, the ECU **50** automatically or with the assistance or direction of the surgeon, e.g., via input signals (not shown), assigns tracking points to at least one coinciding pixel within the stereo image pair **38P**. Within the scope of the method **70**, this action may entail identifying a particular coinciding pixel in each image of the stereo image pair **38P**. Or, if a single coinciding pixel provides insufficient resolution for tracking, a defined pixel cluster of such coinciding pixels may be assigned. The method **70** then proceeds to logic block **B76**.

(30) At logic block **B76** of FIG. **6**, the ECU **50** shown in FIG. **3** next quantifies retinal traction using the collected stereo image data **38**. As part of logic block **B76**, the ECU **50** may automatically perform the above-noted DIC process to the stereo image pair(s) **38P** in the stereo image data **38** in order to ascertain relative motion of the tracking points assigned as part of logic block **B74**. As understood in the art, DIC is an optical technique used in the digital image processing arts to quantify static or dynamic deformation, contour, strain, vibration, and other displacements in an imaged subject. In the present instance, the imaged subject is the retina **25**, the

ILM **31** of FIG. **2**, and the surrounding intraocular tissue, with correlation applied to the identified tracking points from logic block **B74**.

(31) Also as part of logic block **B76**, the ECU **50** also associates the relative motion of such tracking points with retinal traction, as quantified in logic block **B74**. For example, the ECU **50** may reference the traction map **55** shown schematically in FIG. **3**, e.g., as a lookup table indexed by relative motion of the tracking points. A given relative motion value may correspond, for example, to a value referred to herein as a numeric traction quotient, with such a value being indicative of the magnitude of the retinal traction. Such a numeric traction quotient could be normalized in a possible embodiment, with 0 corresponding to no traction on the retina **25** and 1 corresponding to a maximum amount traction. Non-normalized embodiments may be used in the alternative, as may be the actual corresponding traction values.

(32) Motion of the patient eye **30** may occur at times due to patient motion or external forces. For instance, a patient may move during surgery, whether due to the patient's own volition or in response to the surgeon bumping the patient and/or the platform **14** of FIG. **1**, or otherwise. The resulting motion is referred to in the art as free-body or solid-body motion. With respect to such motion, the relative distance between two coinciding image pixels or tracking points in the stereo image pair **38P** remains the same under motion. The relative motion considered herein for the purpose of quantifying retinal traction thus excludes such baseline solid-body motion, e.g., by applying a solid-body motion filter. This enables the ECU **50** to account for solid-body motion of the retina **25** as part of the DIC process occurring within the scope of logic block **B76**. The method **70** proceeds to logic block **B78** once the ECU **50** has determined the numeric traction quotient.

(33) At logic block **B78** of FIG. **6**, the ECU **50** next compares the numeric traction quotient from logic block **B76** to a calibrated traction threshold (“Is Traction $\leq$ CAL?”), or to multiple such traction thresholds corresponding to different regions or zones of the retina **25** as described above. The method **70** repeats logic block **B72** when none of the calibrated traction thresholds are exceeded, i.e., when retinal traction is less than or equal to the above-noted calibrated traction threshold(s). The method **70** proceeds in the alternative to logic block **B80** when the ECU **50** affirmatively determines that one or more of the calibrated traction thresholds have been exceeded.

(34) Logic block **B80** involves executing a control action based on the numeric traction quotient, with the control action being indicative of the retinal traction. For example, the ECU **50** could, in response to the numeric traction quotient exceeding a calibrated traction threshold, activate the indicator device **20A** and/or **20B** in a suitable manner. In embodiments in which the multi-axis surgical robot **60** of FIG. **3** is used, the control action could include transmitting motion control signals to the surgical robot **60**, and in particular to one or more revolute joints thereof as understood in the art, to change a position for force offloading, or in possible autonomous embodiments to change a scraping force, pulling/tugging force, and/or other value as needed in response to the numeric traction quotient.

(35) As part of logic block **B80**, the ECU **50** may consider the magnitude by which a given traction threshold was exceeded in logic block **B78** when determining which of many possible control actions the ECU **50** should execute. That is, the control action could be commensurate with the magnitude of a difference between the current level of retinal traction and the exceeded traction threshold, with the ECU **50** possibly escalating the corresponding alarms as the magnitude increases. Executing the control action may optionally include adjusting a setting of the one or more display screens **20** and/or **200** of FIG. **1** when the numeric traction quotient exceeds a calibrated traction threshold.

(36) An illustrative example includes establishing a corresponding traction threshold for the representative zones **Z1** and **Z2** of FIG. **4**. The ECU **50** could display a color-coded version of the heat map **45** of FIG. **4** via the display screens **20** and/or **200** of FIG. **1**, which would enable the surgeon to discern at a glance whether or not too much traction is being applied in a particular zone relative to another. In an embodiment, the ECU **50** could automatically adjust the color and/or

brightness of “hotter” zones as retinal traction in a given zone progressively increases, such as by gradually coloring the zone from yellow to red as retinal traction increases. Upon crossing a traction threshold for a given zone, the ECU **50** could activate the indicator device **20B** of FIG. **3** in a particular complementary manner, such as by sounding a warning tone, vibrating, displaying a warning message, etc. In lieu of or in addition to such a warning message, the ECU **50** could prompt the surgeon to use a different surgical tool **34**, such as by recommending use of a looped scraper instead of forceps, etc.

(37) By using the ECU **50** of FIGS. **1** and **3**, the intuitive heat map **45** of FIG. **4**, and the method **70** shown in FIG. **6**, a surgeon performing the ophthalmic procedure **13** shown in FIG. **3** is made aware, in an intuitive and possibly localized manner if so desired, of the actual level of traction being applied to the retina **25**. Because alarms are not triggered unless and until a given threshold has been exceeded, the present approach is minimally intrusive and readily customizable via the calibratable inputs **58** (arrow CAL of FIG. **3**) to meet the preferences of a given surgeon.

(38) In terms of such optional customization, it is recognized herein that surgical results are highly dependent upon the individual skillset and techniques employed by a given surgeon. To that end, the ECU **50** of the present disclosure may be configured to present a range of threshold sensitivity options, possibly including using default settings in which a calibrated set of traction thresholds are used for all patients. In some implementations, however, a surgeon may wish to depart from such default settings to properly account for surgical preferences, or to account for different patient-specific/variable parameters such as age, sex, health status such as diabetes, glaucoma, hypertension, etc., previous injuries, diseases, and/or surgeries, past surgical results, and the like. For example, a surgeon in some approaches may be prompted via the ECU **50** to answer a set of questions regarding the patient, including any or all of the above example parameters, with the ECU **50** thereafter recommending or automatically selecting corresponding thresholds to apply in conjunction with the traction map **55**.

(39) With respect to threshold adaptation, in addition to customizing use of the method **70** for a given patient and/or surgeon, the traction thresholds may be adjusted over time using historical data indicative of past surgical results. By way of example, it may be determined over time, using historical results of multiple surgeries over many days, months, or years, that particular control actions associated with exceeding one or more traction thresholds may have been unnecessary, or were premature. In such an instance, the ECU **50** may selectively increase the traction thresholds during subsequent surgeries to permit the surgeon, using the surgeon's professional discretion, to employ higher levels of retinal traction when needed, without fear of damaging the ILM **31** or the retina **25**. Alternatively, when historical data/results indicate that the applied traction thresholds are too low, perhaps resulting in damage or unsatisfactory results where none were expected, the ECU **50** may decrease the traction thresholds.

(40) Those skilled in the art will recognize that the method **70** may be implemented as computer-readable instructions, recorded in memory **54** of FIG. **3** or in a separate memory location, with execution of the instructions by the CPU **52** allowing the CPU **52** to quantify retinal traction on the retina **25** of the patient eye **30** shown schematically in FIG. **3**. That is, execution of the instructions embodying the method **70** causes a system constructed of the CPU **52** and memory **54** to receive the stereo image data **38** from the stereo camera **16** of FIG. **3** during the ophthalmic procedure **13**, and to assign tracking points within the stereo image pair **38P**.

(41) Execution of the instructions also causes the CPU **52** to automatically perform the above-described DIC process using the stereo image pair **38P** to thereby ascertain relative motion of the tracking points, e.g., as shown in FIG. **5**, and to associate relative motion of the tracking points with the retinal traction, using the traction map **55** of FIG. **3**, as a numeric traction quotient indicative of magnitude of the retinal traction. The CPU **52**, alone or using other associated hardware, then communicates the control signal (arrow CC.sub.20A) to the indicator device **20A** when the numeric traction quotient exceeds one or more calibrated traction thresholds. For

instance, the control signal (arrow CC.sub.20A) could be configured to display the above-described color-coded heat map **45** shown in FIG. **4**, thereby intuitively displaying the retinal traction via the display screen **20** and/or **200** of FIG. **1**.

(42) The detailed description and the drawings are supportive and descriptive of the disclosure, but the scope of the disclosure is defined solely by the claims. While some of the best modes and other embodiments for carrying out the claimed disclosure have been described in detail, various alternative designs and embodiments exist for practicing the disclosure defined in the appended claims.

(43) Furthermore, the embodiments shown in the drawings or the characteristics of various embodiments mentioned in the present description are not necessarily to be understood as embodiments independent of each other. Rather, it is possible that each of the characteristics described in one of the examples of an embodiment can be combined with one or a plurality of other desired characteristics from other embodiments, resulting in other embodiments not described in words or by reference to the drawings. Accordingly, such other embodiments fall within the framework of the scope of the appended claims.

## Claims

1. A tracking system for quantifying retinal traction during an ophthalmic procedure, the tracking system comprising: a stereo camera configured to collect and output stereo image data of a retina; and an electronic control unit (ECU) in communication with a display screen and the stereo camera, wherein the ECU is configured to: receive the stereo image data from the stereo camera during the ophthalmic procedure, the stereo image data including stereo image pairs; assign tracking points as coinciding pixels of the stereo image pairs; automatically perform a digital image correlation (DIC) process to the stereo image pairs to ascertain relative motion of the tracking points; associate relative motion of the tracking points with the retinal traction, using a traction map of the ECU, as a numeric traction quotient indicative of a magnitude of the retinal traction; and indicate the magnitude of the retinal traction via the display screen in response to the numeric traction quotient.
2. The tracking system of claim 1, wherein the ECU is configured to provide a real-time alert based on the numeric traction quotient.
3. The tracking system of claim 2, wherein the real-time alert includes one or more of an audio alert, a visual alert, or a tactile alert.
4. The tracking system of claim 2, wherein the ECU is configured to provide the real-time alert in response to the numeric traction quotient exceeding a calibrated traction threshold.
5. The tracking system of claim 1, wherein as part of the DIC process, the ECU is configured to apply a solid-body motion filter to the relative motion to account for solid-body motion of the retina.
6. The tracking system of claim 1, wherein the ECU is configured to adjust a setting of the display screen when the numeric traction quotient exceeds a calibrated traction threshold.
7. The tracking system of claim 6, wherein the ECU is configured to adjust the calibrated traction threshold based on historical data indicative of past surgical results.
8. The tracking system of claim 1, wherein the ECU is configured to display a color-coded heat map of the retinal traction via the display screen to indicate the magnitude of the retinal traction.
9. The tracking system of claim 8, wherein the ECU is configured to display the color-coded heat map of the retinal traction on top of a stereo image formed by the stereo image pairs.
10. The tracking system of claim 1, wherein the ECU is configured to assign the tracking points as coinciding pixels of the stereo image pairs in response to surgeon-directed input signals.
11. A method for quantifying retinal traction on a retina of a patient eye during an ophthalmic procedure, the method comprising: receiving, via an electronic control unit (ECU), stereo image

data of the retina from a stereo camera during the ophthalmic procedure, wherein the stereo image data includes stereo image pairs; assigning tracking points as coinciding pixels of the stereo image pairs in response to input signals; automatically performing a digital image correlation (DIC) process, via the ECU, using the stereo image pairs to thereby ascertain relative motion of the tracking points; associating relative motion of the tracking points with the retinal traction, using a traction map of the ECU, to thereby determine a numeric traction quotient indicative of magnitude of the retinal traction as a normalized value; and indicating the magnitude of the retinal traction via a display screen in response to the normalized value.

12. The method of claim 11, wherein automatically performing the DIC process includes applying a solid-body motion filter to the relative motion to thereby account for solid-body motion of the retina.

13. The method of claim 11, wherein indicating the magnitude of the retinal traction via a display screen includes displaying a color-coded heat map of the retinal traction on top of a stereo image formed from the stereo image pairs.

14. The method of claim 11, further comprising: providing a real-time audio, visual, and/or tactile alert via the ECU when the numeric traction quotient exceeds a calibrated traction threshold.

15. The method of claim 14, further comprising: adjusting the calibrated traction threshold via the ECU based on historical data indicative of past surgical results.

16. The method of claim 14, further comprising: receiving calibratable inputs via the ECU; and adjusting the calibrated traction threshold based on the calibratable inputs.

17. The method of claim 14, wherein the calibrated traction threshold includes a plurality of calibrated traction thresholds each corresponding to a different zone or region of the retina.

18. The method of claim 11, wherein assigning the tracking points as coinciding pixels of the stereo image pairs in response to input signals occurs at least in part in response to surgeon-directed input signals during the ophthalmic procedure.

19. A system for quantifying retinal traction on a retina of a patient eye, comprising: a central processing unit (CPU); and computer-readable media on which is recorded instructions, wherein execution of the instructions by the CPU causes the CPU to: receive stereo image data from a stereo camera during an ophthalmic procedure, the stereo image data including stereo image pairs; assign tracking points as coinciding pixels within the stereo image pairs; automatically perform a digital image correlation (DIC) process using the stereo image pairs to thereby ascertain relative motion of the tracking points; associate relative motion of the tracking points with the retinal traction, using a traction map, as a numeric traction quotient indicative of magnitude of the retinal traction; and communicate a control signal to one or more display screens when the numeric traction quotient exceeds one or more calibrated traction thresholds to thereby cause the one or more display screens to display a color-coded heat map of the retinal traction.

20. The system of claim 19, wherein the execution of the instructions by the CPU causes the CPU to adapt the one or more calibrated traction thresholds over time in response to historical data indicative of past surgical results.

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